THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE UNIT TEST SET 2

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The 15th term of the AP $5, 8, 11, 14, \ldots$ is: 1

(a) 48 **(b)** 49 **(c)** 47 **(d)** 46

2. In a cyclic quadrilateral $ABCD$, $\angle A = 55^\circ$ and the side BC is produced to E. Then $\angle DCE =$ 1

(a) 65° **(b)** 55° **(c)** 45° **(d)** 75°

3. The area of a rhombus with diagonals 14 cm and 10 cm is (cm^2): 1

(a) 69 **(b)** 72 **(c)** 71 **(d)** 70

4. A bag has 4 red, 3 black and 5 white balls. The probability of drawing a ball that is not black is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{3}{5}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{2}$

5. The sum of the first 10 terms of the AP $3, 7, 11, 15, \ldots$ is: 1

(a) 230 **(b)** 220 **(c)** 200 **(d)** 210

6. In triangle OAB, $OA = OB$ and $\angle AOB = 80^\circ$ (O is the centre). Then $\angle OAB =$ 1

(a) 70° **(b)** 50° **(c)** 60° **(d)** 40°

7. The area of a cyclic quadrilateral with sides 4 cm, 5 cm, 7 cm and 10 cm is (cm^2): 1

(a) 37 **(b)** 35 **(c)** 38 **(d)** 36

8. If $P(E) = 0.3$, then $P(\text{not } E) \times 100$ equals: 1

(a) 69 **(b)** 70 **(c)** 71 **(d)** 72

9. The 5th term of the GP $2, 6, 18, \ldots$ is: 1

(a) 164 **(b)** 160 **(c)** 163 **(d)** 162

10. Equal chords of a circle are ___ from the centre. 1

(a) equidistant **(b)** on the same line

(c) at distances in ratio $1 : 2$ **(d)** at different distances

11. The length of an arc of angle 120° in a circle of radius 21 cm (take $\pi = \frac{22}{7}$) is (cm): 1

(a) 44 **(b)** 43 **(c)** 45 **(d)** 46

12. A fair die is thrown once. The probability of getting a composite number is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{3}$

13. The number of terms in the AP $3, 7, 11, 15, \ldots$ ending at 83 is: 1

(a) 21 **(b)** 22 **(c)** 20 **(d)** 23

14. A chord is at distance 24 cm from the centre of a circle of radius 25 cm. Its length (in cm) is: 1

(a) 15 **(b)** 13 **(c)** 14 **(d)** 16

15. A square has the same perimeter as a circle of circumference 88 cm. The side of the square (cm) is: 1

(a) 24 **(b)** 23 **(c)** 22 **(d)** 21

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A wheel of radius 28 cm covers a distance of 4400 cm. How many revolutions does it make (take $\pi = \frac{22}{7}$)? 2

17. A letter is chosen at random from the word MATHEMATICS. Find the probability that it is a consonant. 2

18. Show that $8, 13, 18, 23, \ldots$ is an AP and find its 12th term. 2

19. $ABCD$ is a cyclic quadrilateral and BC is produced to E. If $\angle A = 62^\circ$, find $\angle DCE$. 2

20. Find the length of an arc of 120° in a circle of radius 21 cm (take $\pi = \frac{22}{7}$). 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the sum of the first 15 terms of the AP $6, 11, 16, 21, \ldots$. 3

22. A chord of length 30 cm is 8 cm from the centre of a circle. Find the radius and the area of the right triangle formed by the radius, the perpendicular from the centre and half the chord. 3

23. Find the area of a parallelogram with base 14 cm and height 9 cm, and the base of a triangle of height 18 cm having the same area. 3

24. Two coins are tossed 100 times: two heads occurred 24 times, one head 52 times and no head 24 times. Find the experimental probability of each of the three events. 3

OR

Find the common ratio of the GP $3, 12, 48, \ldots$ and its 5th term. 3

25. In triangle OAB with O the centre of a circle, $\angle AOB = 90^\circ$. Find $\angle OAB$ and the angle subtended by AB at a point of the major arc. 3

OR

Find the area of the triangle with sides 10 cm, 17 cm and 21 cm, and the length of the altitude on the side of 21 cm. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. $ABCD$ is a cyclic quadrilateral with $\angle A = 75^\circ$, $\angle B = 88^\circ$ and side BC is produced to E, side AB to F. Find $\angle C$, $\angle D$, $\angle DCE$ and $\angle DAF$. 5

OR

A rectangular lawn 30 m $\times$ 20 m has a path 2 m wide all around it, outside the lawn. Find the area of the path. 5

27. A letter is chosen at random from the word MISSISSIPPI. Find the probability that it is 5

(i) a vowel,

(ii) the letter S,

(iii) a letter from the first half of the alphabet (A to M).

OR

Rohan saves ₹300 in the first month and ₹50 more each month than in the previous month. 5

(i) How much does he save in month 12?

(ii) How much does he save in 12 months in all?

END OF QUESTION PAPER **Total: 50 Marks**

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